21 May 2002

TO: Pre–Installation Tab 088

SUBJECT: **Direction 2304880, Signa Twin*Speed* Upgrade Pre–Installation, Rev 1**

Enclosed is *Direction 2304880, Signa TwinSpeed Upgrade Pre–Installation,* Rev 1 revision. Table 1 below summarizes the major information incorporated in this revision. Changed content on individual pages is indicated by revision bar in the left hand margin.

TABLE 1
DIRECTION 2304880 REVISION 1 MAJOR CHANGES INCORPORATED

TAB	MAJOR INFORMATION CHANGES
1 SYSTEM CONFIGURATION	• Added Signa Twin*Speed* with EXCITE Technology upgrade catalogs
2 ROOM LAYOUTS	• Added Signa Twin*Speed* with EXCITE Technology upgrade equipment information • Update TSCC & TGWC for revised E&W information • Added Network connection and telephone lines option to system connectivity requirements • Updated installed weight of Magnet, Enclosure, Body Coil, & Cryogens • Updated TAC Cabinet dimensions drawing
4 SITE ENVIRONMENT	• Added Outdoor TSCC & TGWC Operating Environment temperature & humidity specifications • Updated Water Cooled TSCC & TGWC water cooling requirements • Added System Cabinet with EXCITE Technology alarm device information
5 POWER REQUIREMENTS	• Revised Voltage Transient specificiation • Added NEC 2002 Article references
6 INTERCONNECT DATA	• Added Signa Twin*Speed* with EXCITE Technology information • Updated miscellanous references
7 RF SHIELDED ROOM	• Corrected Table Of Contents typographical error
9 PREINSTALLATION CHECKLIST	• Updated SOI Request Form web address • Updated references to Network connection and telephone lines option to system connectivity • Revised checklist item to inform outside emergency personnel precautions to take in event of emergencies in advance of magnet ramp–up
10 TOOLS & TEST EQUIPMENT	• Added new & deleted obsolete test equipment & part numbers
APPENDIX B	• Added Ellis & Watts “Prerequisites for Scheduled Performance Verification of Chillers For Signa Twin*Speed*” document.

UPDATE INSTRUCTIONS FOR YOUR DIRECTION 2294003 REV 1

Refer to Table 2 for instructions to properly update your Direction 2294003 Rev 0 with the enclosed Revision 1 pages.

TABLE 2
DIRECTION 2304880 REVISION 1 UPDATE INSTRUCTIONS

TAB	REMOVE PAGE(S)	INSERT PAGE(S)
Front Matter	Title Page	Title Page
	A	A
	i to v	i to v
1 SYSTEM CONFIGURATION	1–1 to 1–33 (entire section)	1–1 to 1–58 (entire section)
2 ROOM LAYOUTS	2–1 to 2–62 (entire section)	2–1 to 2–68 (entire section)
4 SITE ENVIRONMENT	4–1 to 4–19 (entire section)	4–1 to 4–21 (entire section)
5 POWER REQUIREMENTS	5–1 to 5–17 (entire section)	5–1 to 5–17 (entire section)
6 INTERCONNECT DATA	6–1 to 6–50 (entire section)	6–1 to 6–69 (entire section)
7 RF SHIELDED ROOM	7–1/2	7–1/2
9 PREINSTALLATION CHECKLIST	9–1 to 9–7 (entire section)	9–1 to 9–7 (entire section)
10 TOOLS & TEST EQUIPMENT	10–1 to 1–10 (entire section)	10–1 to 1–10 (entire section)
APPENDIX B		Tab B Ellis & Watts “Prerequisites for Scheduled Performance Verification of Chillers For Signa Twin*Speed*” document

Debra C. Balis
MR SERVICE ENGINEERING

GE Medical Systems

Technical Publications

Direction 2304880

Revision 1

Signa® 1.5T Twin*Speed* Upgrade

Pre–Installation

Architectural/Installation Planning Use Only

Operating Documentation

GE Medical Systems

GE Medical Systems: Telex 3797371
P.O. Box 414, Milwaukee, Wisconsin 53201 U.S.A.
(Asia, Pacific, Latin America, North America)

GE Medical Systems – Europe: Telex 261794
Shortlands, Hammersmith, London W6 8BX U. K.

WARNING

- **THIS SERVICE MANUAL IS AVAILABLE IN ENGLISH ONLY.**
- **IF A CUSTOMER'S SERVICE PROVIDER REQUIRES A LANGUAGE OTHER THAN ENGLISH, IT IS THE CUSTOMER'S RESPONSIBILITY TO PROVIDE TRANSLATION SERVICES.**
- **DO NOT ATTEMPT TO SERVICE THE EQUIPMENT UNLESS THIS SERVICE MANUAL HAS BEEN CONSULTED AND IS UNDERSTOOD.**
- **FAILURE TO HEED THIS WARNING MAY RESULT IN INJURY TO THE SERVICE PROVIDER, OPERATOR OR PATIENT FROM ELECTRIC SHOCK, MECHANICAL OR OTHER HAZARDS.**

AVERTISSEMENT

- **CE MANUEL DE MAINTENANCE N'EST DISPONIBLE QU'EN ANGLAIS.**
- **SI LE TECHNICIEN DU CLIENT A BESOIN DE CE MANUEL DANS UNE AUTRE LANGUE QUE L'ANGLAIS, C'EST AU CLIENT QU'IL INCOMBE DE LE FAIRE TRADUIRE.**
- **NE PAS TENTER D'INTERVENTION SUR LES ÉQUIPEMENTS TANT QUE LE MANUEL SERVICE N'A PAS ÉTÉ CONSULTÉ ET COMPRIS.**
- **LE NON-RESPECT DE CET AVERTISSEMENT PEUT ENTRAÎNER CHEZ LE TECHNICIEN, L'OPÉRATEUR OU LE PATIENT DES BLESSURES DUES À DES DANGERS ÉLECTRIQUES, MÉCANIQUES OU AUTRES.**

WARNUNG

- **DIESES KUNDENDIENST–HANDBUCH EXISTIERT NUR IN ENGLISCHER SPRACHE.**
- **FALLS EIN FREMDER KUNDENDIENST EINE ANDERE SPRACHE BENÖTIGT, IST ES AUFGABE DES KUNDEN FÜR EINE ENTSPRECHENDE ÜBERSETZUNG ZU SORGEN.**
- **VERSUCHEN SIE NICHT, DAS GERÄT ZU REPARIEREN, BEVOR DIESES KUNDENDIENST–HANDBUCH NICHT ZU RATE GEZOGEN UND VERSTANDEN WURDE.**
- **WIRD DIESE WARNUNG NICHT BEACHTET, SO KANN ES ZU VERLETZUNGEN DES KUNDENDIENSTTECHNIKERS, DES BEDIENERS ODER DES PATIENTEN DURCH ELEKTRISCHE SCHLÄGE, MECHANISCHE ODER SONSTIGE GEFAHREN KOMMEN.**

AVISO

- **ESTE MANUAL DE SERVICIO SÓLO EXISTE EN INGLÉS.**
- **SI ALGÚN PROVEEDOR DE SERVICIOS AJENO A GEMS SOLICITA UN IDIOMA QUE NO SEA EL INGLÉS, ES RESPONSABILIDAD DEL CLIENTE OFRECER UN SERVICIO DE TRADUCCIÓN.**
- **NO SE DEBERÁ DAR SERVICIO TÉCNICO AL EQUIPO, SIN HABER CONSULTADO Y COMPRENDIDO ESTE MANUAL DE SERVICIO.**
- **LA NO OBSERVANCIA DEL PRESENTE AVISO PUEDE DAR LUGAR A QUE EL PROVEEDOR DE SERVICIOS, EL OPERADOR O EL PACIENTE SUFRAN LESIONES PROVOCADAS POR CAUSAS ELÉCTRICAS, MECÁNICAS O DE OTRA NATURALEZA.**

ATENÇÃO

- **ESTE MANUAL DE ASSISTÊNCIA TÉCNICA SÓ SE ENCONTRA DISPONÍVEL EM INGLÊS.**
- **SE QUALQUER OUTRO SERVIÇO DE ASSISTÊNCIA TÉCNICA, QUE NÃO A GEMS, SOLICITAR ESTES MANUAIS NOUTRO IDIOMA, É DA RESPONSABILIDADE DO CLIENTE FORNECER OS SERVIÇOS DE TRADUÇÃO.**
- **NÃO TENTE REPARAR O EQUIPAMENTO SEM TER CONSULTADO E COMPREENDIDO ESTE MANUAL DE ASSISTÊNCIA TÉCNICA.**
- **O NÃO CUMPRIMENTO DESTE AVISO PODE POR EM PERIGO A SEGURANÇA DO TÉCNICO, OPERADOR OU PACIENTE DEVIDO A‘ CHOQUES ELÉTRICOS, MECÂNICOS OU OUTROS.**

AVVERTENZA

- **IL PRESENTE MANUALE DI MANUTENZIONE È DISPONIBILE SOLTANTO IN INGLESE.**
- **SE UN ADDETTO ALLA MANUTENZIONE ESTERNO ALLA GEMS RICHIEDE IL MANUALE IN UNA LINGUA DIVERSA, IL CLIENTE È TENUTO A PROVVEDERE DIRETTAMENTE ALLA TRADUZIONE.**
- **SI PROCEDA ALLA MANUTENZIONE DELL’APPARECCHIATURA SOLO DOPO AVER CONSULTATO IL PRESENTE MANUALE ED AVERNE COMPRESO IL CONTENUTO.**
- **NON TENERE CONTO DELLA PRESENTE AVVERTENZA POTREBBE FAR COMPIERE OPERAZIONI DA CUI DERIVINO LESIONI ALL’ADDETTO ALLA MANUTENZIONE, ALL’UTILIZZATORE ED AL PAZIENTE PER FOLGORAZIONE ELETTRICA, PER URTI MECCANICI OD ALTRI RISCHI.**

警告

・このサービスマニュアルには英語版しかありません。
・ＧＥＭＳ以外でサービスを担当される業者が英語以外の言語を要求される場合、翻訳作業はその業者の責任で行うものとさせていただきます。
・このサービスマニュアルを熟読し理解せずに、装置のサービスを行わないで下さい。
・この警告に従わない場合、サービスを担当される方、操作員あるいは患者さんが、感電や機械的又はその他の危険により負傷する可能性があります。

注意:

- 本维修手册仅存有英文本。
- 非 GEMS 公司的维修员要求非英文本的维修手册时，客户需自行负责翻译。
- 未详细阅读和完全了解本手册之前，不得进行维修。
- 忽略本注意事项会对维修员，操作员或病人造成触电，机械伤害或其他伤害。

REVISION HISTORY

REV	DATE	PRIMARY REASON FOR CHANGE
0	Aug. 31, 2001	Initial document.
1	May 21, 2002	Added Signa TwinSpeed with EXCITE Technology system upgrade information throughout the document. Other major changes listed by section include the following **Sec 2:** Update TSCC & TGWC for revised E&W information; Added Network connection and telephone lines option to system connectivity requirements; Updated installed weight of Magnet, Enclosure, Body Coil, & Cryogens; Updated TAC Cabinet dimensions drawing. **Sec 4:** Added Outdoor TSCC & TGWC Operating Environment temperature & humidity specifications; Updated Water Cooled TSCC & TGWC water cooling requirements. **Sec 5:** Revised Voltage Transient specificiation; Added NEC 2002 Article references. **Sec 6:** Updated miscellanous references. **Sec 7:** Corrected Table Of Contents typographical error. **Sec 9:** Updated SOI Request Form web address; Updated references to Network connection and telephone lines option to system connectivity. **Sec 10:** Added new & deleted obsolete test equipment & part numbers; Revised checklist item to inform outside emergency personnel precautions to take in event of emergencies in advance of magnet ramp–up. **Appendix B:** Added Tab B and Ellis & Watts "Prerequisites for Scheduled Performance Verification of Chillers For Signa Twin*Speed*" document.

PAGE	REV
Title Page	1
GE Logo	–
Direction 2128126	0*
A	1
i to iv	1
v	0
SYSTEM CONFIG.	
Tab MR251	–
1–1	1
1–2	blank
1–3 to 1–58	1
ROOM LAYOUTS	
Tab MR252	–
2–1	1
2–2	blank
2–3 to 2–4	1
2–5 to 2–6	0
2–7	1
2–8 to 2–14	0
2–15	1
2–16	0
2–17 to 2–18	1
2–19	0
2–20 to 2–68	1
MAGNETIC FIELD CONSID.	
Tab MR253	–
3–1	0
SITE ENVIRONMENT	
Tab MR254	–
4–1	1
4–2	blank
4–3	1
4–4	0
4–5	1
4–6 to 4–7	0
4–8 to 4–21	1
POWER REQUIREMENTS	
Tab MR255	–
5–1	0
5–2	blank
5–3	0
5–4	1
5–5	0
5–6	1
5–7 to 5–8	0
5–9	1
5–10	0
5–11	1
5–12 to 5–16	0
5–17	1
INTERCONNECT DATA	
Tab MR256	–
6–1 to 6–4	1
6–5	0
6–6 to 6–69	1
RF SHIELDED ROOM	
Tab MR257	–
7–1	1
7–2	blank
7–3 to 7–4	0
7–5	1
7–6	0
SHIPPING & DELIVERY	
Tab MR258	–
8–1	0
8–2	blank
8–3 to 8–14	0
PREINSTALL. CHECKLIST	
Tab MR259	–
9–1	0
9–2	blank
9–3 to 9–5	1
9–6	0
9–7	1
TOOLS & TEST EQUIPMENT	
Tab MR260	–
10–1	0
10–2	blank
10–3 to 10–5	0
10–6 to 10–8	1
10–9	0
10–10	1
A	
Tab MR301	–
A–1 to A–7	0
B	
Tab MR302	–
E&W schedule	02/14/02*

* This revision number/letter corresponds to the indicated document's revision control system.

TABLE OF CONTENTS

SECTION **PAGE**

INTRODUCTION

This document contains the physical, magnetic, cryogenic, plumbing and electrical data necessary for planning and preparing a site for a Signa 1.5T TwinSpeed system upgrade. "Preinstallation work" is done to prepare the customer's premises for the installation of the upgrade sold. It is the responsibility of the purchaser to arrange for performance of this work. Such work includes:

- Installation of the electrical conduit, junction boxes, ducts, surface raceways, outlets and line safety switches.
- Installation of wires not supplied by GE Medical System such as facility input power line to the Main Disconnect Panel and Power Distribution Unit. If new wires are installed then the electrical contractor shall ring identify and tag all wires at both ends. Color–coded wires are recommended for easier identification. Wires shall be continuous without splices. Insulation on all equipment ground wires must be green with a yellow stripe.
- Installation of non–electrical lines such as: water plumbing, helium venting systems and air conditioning equipment. Also, installation of recommended air, vacuum and oxygen lines into the Magnet Room. All lines must be clearly labeled.
- Site renovation.
- Installation of structural reinforcements as required.

All work MUST be in compliance with national and local building and safety codes.

All site plans and preliminary concepts should be reviewed by GE Medical Systems MR Siting group prior to construction.

Unless specifically mentioned, GE Medical Systems does not provide or install: the facility input power lines to the Main Disconnect Panel and Power Distribution Unit or the power lines required in the Magnet Room, the Main Disconnect Panel option, nor raised flooring, conduit, junction boxes, ducts, plumbing, water treatment equipment, cryogenic venting outside the Magnet Room, non standard cryogenic venting inside the Magnet Room, or the RF shielded room specified in this document.

All electrical installations that are preliminary to positioning of the equipment at the site prepared for the equipment shall be performed by licensed electrical contractors. In addition, electrical feeds into the Main Disconnect Panel and Power Distribution Unit shall be performed by licensed electrical contractors. Other connections between pieces of electrical equipment, calibrations, and testing shall be performed by qualified GE Medical Systems personnel or third–party service companies with equivalent training. The products involved (and the accompanying electrical installations) are highly sophisticated, and special engineering competence is required. In performing all electrical work on these products, GE will use its own specially trained field engineers. All of GE's electrical work on these products will comply with the requirements of the applicable electrical codes. The purchaser of GE equipment shall only utilize qualified personnel (i.e. GE's field engineers, personnel of third–party service companies with equivalent training, or licensed electricians) to perform electrical servicing on the equipment.

Pre–installation information is continually changing due to the evolution of the product. GE will make every reasonable effort to maintain accuracy of the pre–installation information.